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EXPERIMENT 15

ENDPOINT SECURITY MONITORING

AIM: To understand the basics of endpoint security and how devices are protected from cyber threats.

PROCEDURE:

1. Learn what endpoint devices are (like laptops, mobiles, desktops).
2. Understand common threats that affect these devices.
3. Study protection methods like antivirus and firewall.
4. Learn how monitoring tools detect suspicious activities.
5. Understand the importance of updating and securing devices regularly.

1.

Task 1 Room Introduction

In this room, we will introduce the fundamentals of endpoint security monitoring, essential tools, and high-level methodology. This room gives an overview of determining a malicious activity from an endpoint and mapping its related events.

To start with, we will tackle the following topics to build a stepping stone on how to deal with Endpoint Security Monitoring.

- Endpoint Security Fundamentals
- Endpoint Logging and Monitoring
- Endpoint Log Analysis

At the end of this room, we will have a threat simulation wherein you need to investigate and remediate the infected machines. This activity may require you first to understand the fundamentals of endpoint security monitoring to complete it.

Now, let's deep-dive into the basics of Endpoint Security!

Answer the questions below

I have read the introduction task.

No answer needed

✓ Correct Answer

2.

Task 2 ✔ Endpoint Security Fundamentals

Core Windows Processes

Before we deal with learning how to deep-dive into endpoint logs, we need first to learn the fundamentals of how the Windows Operating System works. Without prior knowledge, differentiating an outlier from a haystack of events could be problematic.

To learn more about Core Windows Processes, a built-in Windows tool named Task Manager may aid us in understanding the underlying processes inside a Windows machine.

Task Manager is a built-in GUI-based Windows utility that allows users to see what is running on the Windows system. It also provides information on resource usage, such as how much each process utilizes CPU and memory. When a program is not responding, the Task Manager is used to terminate the process.

Task Manager

File Options View

Processes Performance Users Details Services

Name	PID	Status	User name	Image path name	Command line
System interrupts	-	Running	SYSTEM		
System Idle Process	0	Running	SYSTEM		
System	4	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
Registry	88	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
smss.exe	280	Running	SYSTEM	C:\Windows\System32\smss.exe	
svchost.exe	384	Running	LOCAL SERVICE	C:\Windows\System32\svchost.exe	C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p

A Task Manager provides some of the Core Windows Processes running in the background. Below is a summary of running processes that are considered normal behaviour.

Note: **">"** symbol represents a parent-child relationship. `System (Parent) > smss.exe (Child)`

- System
- System > smss.exe
- csrss.exe
- wininit.exe
- wininit.exe > services.exe
- wininit.exe > services.exe > svchost.exe
- lsass.exe
- winlogon.exe
- explorer.exe

In addition, the processes with no depiction of a parent-child relationship should not have a Parent Process under normal circumstances, except for the System process, which should only have **System Idle Process (0)** as its parent process.

You may refer to the Core Windows Processes Room to learn more about this topic.

Answer the questions below

What is the normal parent process of services.exe?

wininit.exe ✓ Correct Answer

What is the name of the network utility tool introduced in this task?

TCPView ✓ Correct Answer

3.

Task 3 Endpoint Logging and Monitoring

From the previous task, we have learned basic knowledge about the Windows Operating system in terms of baseline processes and essential tools to analyze events and artefacts running on the machine. However, this only limits us from observing real-time events. With this, we will introduce the importance of endpoint logging, which enables us to audit significant events across different endpoints, collect and aggregate them for searching capabilities, and better automate the detection of anomalies.

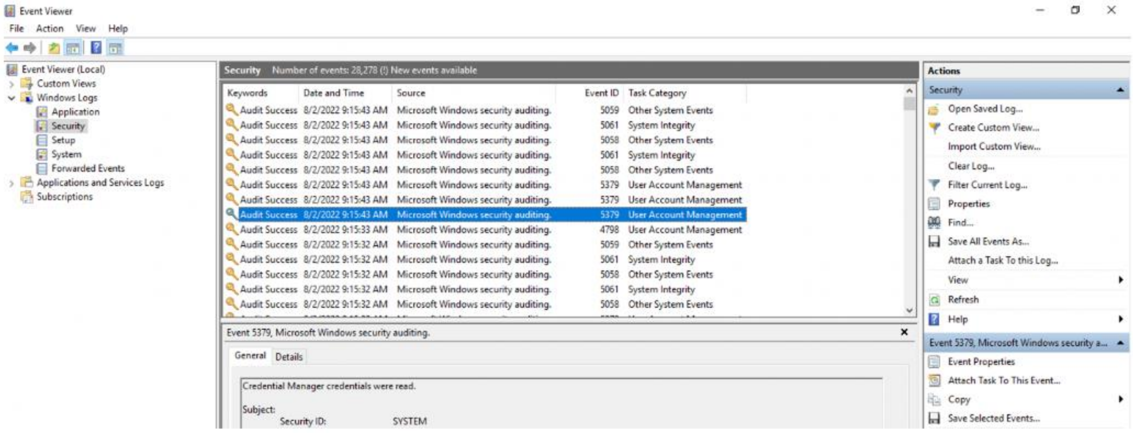
Windows Event Logs

The Windows Event Logs are not text files that can be viewed using a text editor. However, the raw data can be translated into XML using the Windows API. The events in these log files are stored in a proprietary binary format with a .evt or .evtx extension. The log files with the .evtx file extension typically reside in `C:\Windows\System32\winevt\Logs`.

There are three main ways of accessing these event logs within a Windows system:

1. **Event Viewer** (GUI-based application)
2. **Wevtutil.exe** (command-line tool)
3. **Get-WinEvent** (PowerShell cmdlet)

An example image of logs viewed using the **Event Viewer** tool is shown below.



You may refer to the Windows Event Logs Room to learn more about Windows Event Logs.

Answer the questions below

Where do the Windows Event logs (.evtx files) typically reside?

C:\Windows\System32\winevt\Logs ✓ Correct Answer

Provide the command used to enter OSQuery CLI.

osqueryi ✓ Correct Answer

What does EDR mean? Provide the answer in lowercase.

endpoint detection and response ✓ Correct Answer

4.

Task 4 Endpoint Log Analysis

Event Correlation

View Site

Event correlation identifies significant relationships from multiple log sources such as application logs, endpoint logs, and network logs.

Event correlation deals with identifying significant artefacts co-existing from different log sources and connecting each related artefact. For example, a network connection log may exist in various log sources such as Sysmon logs (Event ID 3: Network Connection) and Firewall Logs. The Firewall log may provide the source and destination IP, source and destination port, protocol, and the action taken. In contrast, Sysmon logs may give the process that invoked the network connection and the user running the process.

With this information, we can connect the dots of each artefact from the two data sources:

- Source and Destination IP
- Source and Destination Port
- Action Taken
- Protocol
- Process name
- User Account
- Machine Name

Event correlation can build the puzzle pieces to complete the exact scenario from an investigation.

Baselining

Baselining is the process of knowing what is expected to be normal. In terms of endpoint security monitoring, it requires a vast amount of data-gathering to establish the standard behaviour of user activities, network traffic across infrastructure, and processes running on all machines owned by the organization. Using the baseline as a reference, we can quickly determine the outliers that could threaten the organization.

Below is a sample list of baseline and unusual activities to show the importance of knowing what to expect in your network.

Baseline	**Unusual Activity**
The organization's employees are in London, and the regular working hours are between 9 AM and 6 PM.	A user has authenticated via VPN connecting from Singapore at 3 AM.
A single workstation is assigned to each employee.	A user has attempted to authenticate to multiple workstations.

Answer the questions below

Click on the green View Site button in this task to open the Static Site Lab and start investigating the threat by following the provided instructions.

No answer needed | ✓ Correct Answer

Provide the flag for the simulated investigation activity.

THM{3ndp01nt_s3cur1ty!} | ✓ Correct Answer

5.

Task 5 Conclusion

Congratulations! You have completed the investigation task.

In the simulated threat investigation activity, we have learned the following:

- Having a baseline document aids you in differentiating malicious events from benign ones.
- Event correlation provides a deeper understanding of the concurrent events triggered by the malicious activity.
- Taking note of each significant artefact is crucial in the investigation.
- Other potentially affected assets should be inspected and remediated using the collected malicious artefacts.

In conclusion, we covered the basic concepts of Endpoint Security Monitoring:

- **Endpoint Security Fundamentals** tackled Core Windows Processes and Sysinternals.
- **Endpoint Logging and Monitoring** introduced logging functionalities such as Windows Event Logging and Sysmon and monitoring/investigation tools such as OSQuery and Wazuh .
- **Endpoint Log Analysis** highlighted the importance of having a methodology such as baselining and event correlation.

You are now ready to deep-dive into the Endpoint Security Monitoring Module. To continue this path, you may refer to the list of rooms mentioned in the previous tasks:

- Core Windows Processes
- Sysinternals
- Windows Event Logs
- Sysmon
- OSQuery
- Wazuh

Answer the questions below

I have completed the Introduction to Endpoint Security Monitoring room.

No answer needed

✓ Correct Answer

RESULT: Thus, the basics of endpoint security and device protection were understood successfully.